

NanoSafe Analyzer — Research & Cytotoxicity Report

In Vitro Evaluation of ZnO Nanoparticle Biocompatibility | ISO 10993-5 Compliance Assessment

Report ID: #14 | Generated: 2026-08-17 12:34:30 | Analyst: Researcher

1. EXPERIMENT OVERVIEW

Experiment / Sample Name:	Experiment 1111	Nanoparticle Type:	ZnO
Cell Line Evaluated:	HeLa	Exposure Duration:	N/A
Medical Application:	General Biomedical	ISO 10993-5 Compliance:	PASS — Biocompatible
Lead Researcher:	Researcher	Data Source:	Direct Entry

2. STUDY PARTICIPANT & BIOLOGICAL SAMPLE TRACEABILITY

Biological Sample: Standard in-vitro reference cell culture line. Anonymized laboratory experiment.

3. AVERAGED CYTOTOXICITY & BIOMARKER METRICS

Mean Cell Viability:	93.33%	Mean ZnO Concentration:	96.0 µg/mL
Reactive Oxygen Species (ROS):	5.333333333333333x	LDH Membrane Leakage:	6.666666666666667%
Apoptosis Inducement:	6.0%		

4. MACHINE LEARNING CYTOTOXICITY ASSESSMENT

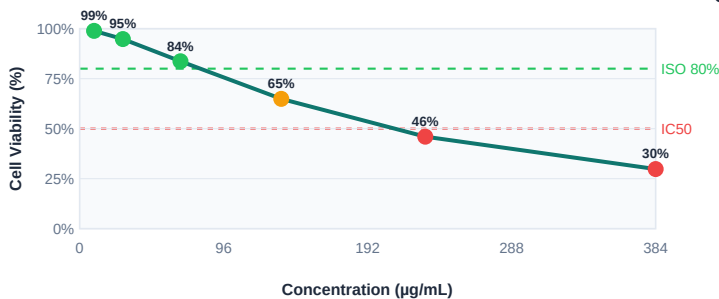
Predicted Toxicity Score:	5.74 / 100	Risk Classification:	Low
Estimated IC50 (4PL Curve):	Not Reached (> Maximum Tested Dose)	Biocompatible Safe Window:	0.0 - 96.0 µg/mL
ML Model Confidence:	98.5%		

5. SCIENTIFIC INTERPRETATION & RECOMMENDATION

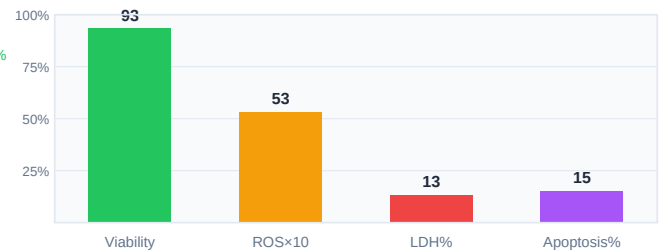
Local Scikit-Learn Ensemble Model evaluated ZnO cytotoxicity on HeLa cells under a 24.0h exposure duration for 'Wound Dressing & Topical Antimicrobial'. Measured parameters: Viability=93.33%, ROS=5.333333333333333, LDH=6.666666666666667%, Apoptosis=6.0%. The offline ML model computed a Toxicity Score of 5.74 out of 100 (Low Risk Zone, ISO 10993-5 status: PASS — Biocompatible). Predicted IC50: 45.39 µg/mL, Safe Biomedical

6. DOSE-RESPONSE CURVE & BIOMARKER PROFILE CHARTS

Graph A: Cell Viability vs. Concentration (Dose-Response Curve)



Graph B: Multi-Biomarker Safety Profile



7. CLINICAL SUGGESTIONS & NEXT STEPS

- SAFE FOR USE:**
Cell viability at 93.3% exceeds ISO 10993-5 threshold of >=80%. The formulation demonstrates acceptable biocompatibility for in-vitro conditions.
- NEXT STEPS:**
Proceed to Phase II in-vivo validation in an appropriate animal model (e.g., murine wound model). Evaluate systemic toxicity and immune response at escalating dose levels.
- REGULATORY PATH:**
Compile cytotoxicity dataset per ISO 10993-5 guidance for medical device regulatory submission. Retain raw data, chain of custody records, and this report for audit compliance.